Problems

1. Different species of crickets have distinct songs, and they use these songs for mate recognition. Researchers crossed two species of Hawaiian crickets (*Laupala paranigra* and *L. kohalensis*) whose songs are distinguished by pulse rate (the number of pulses per second; Shaw et al., *Molecular Ecology* 16, 2007, 2879–2892.) Then, they mapped QTL in the F2 population derived from this cross. Six autosomal QTL were detected. The mean trait values (pulses per second) at the three genotypic classes in the F2 for each QTL are shown in the table below, where P indicates the *L. paranigra* allele and K indicates the *L. kohalensis* allele.

a. Calculate the additive (A) and dominance (D) effects and the D/A ratio for each of the six QTL.

b. Which of these QTL shows the greatest amount of dominance?

c. Which of these has the largest additive effect?

d. The mean pulse rate for *L. kohalensis* is 3.72, and it is 0.71 for *L. paranigra*. Do all six QTL act in the expected direction with the *L. kohalensis* allele conferring a higher pulse rate than the *L. paranigra* allele?

QTL	P/P	P/K	K/K
1	1.54	1.89	2.10
2	1.75	1.87	1.94
3	1.72	1.88	1.92
4	1.70	1.82	2.02
5	1.67	1.80	2.13
6	1.57	1.88	2.19

a .

$$A = \frac{X_{B_2B_2} - X_{B_1B_1}}{2} \quad D = X_{B_1B_2} - \left(\frac{X_{B_2B_2} + X_{B_1B_1}}{2} \right)$$

QTL	P/P	P/K	K/K	A	D	D/A
1	1.54	1.89	2.10	0.28	0.07	0.25
2	1.75	1.87	1.94	0.095	0.025	0.263
3	1.72	1.88	1.92	0.1	0.06	0.6
4	1.70	1.82	2.02	0.16	-0.04	-0.25
5	1.67	1.80	2.13	0.23	-0.1	-0.435
6	1.57	1.88	2.19	0.31	0	0

b. QTL 3

因为 D/A 最高。

c. QTL 6

因为它 additive effect 值最高。

d. 是，因为 A 都大于 0。

Problems

2. QTL mapping and GWA (association) mapping are two different methods used to identify genes that affect complex traits. For each of the following statements, choose whether it applies to QTL mapping, association mapping, or both.

Statement	QTL	GWA	Both
This method requires that the experimenter make crosses between different strains to produce a mapping population.			
This method can scan the entire genome to find QTL for a trait.			
This method can often identify the specific genes that represent the QTL.			
This method may sample a large number of individuals from a random-mating population that has variation for the trait being studied.			
This method typically tests two alleles that differ between the two parents of the mapping population.			

QTL
Both
GWA
GWA
QTL

Problems

3. The *ocular albinism-2* (*OCA2*) gene and the *melanocortin- 1-receptor* (*MC1R*) gene are both involved in melanin metabolism in skin cells in humans. To test whether variation at these genes contributes to sun sensitivity and the associated risk of being afflicted with skin cancer, you perform association analyses. A sample of 1000 people from Iceland were asked to classify themselves as having tanning or burning (nontanning) skin when exposed to the sun. The individuals were also genotyped for a SNP in each gene (rs7495174 and rs1805007). The table shows the number of individuals in each class.

a. What are the frequencies of tanning and burning phenotypes in Iceland?

b. What are the allelic frequencies at each locus (SNP)?

c. Using the $\chi 2$ test (see Chapter 3) and Table 3-1, test the null hypothesis that there is no association between these SNPs and sun-sensitive skin. Does either SNP show evidence for an association?

d. If you find evidence for an association between the gene and the trait, what is the mode of gene action?

e. If the P value is greater than 0.05, does that prove that the gene does not contribute to variation for sun sensitivity? Why?

	OCA2 (rs7495174)			*MC1R* (rs1805007)		
	A/A	A/G	G/G	C/C	C/T	T/T
Burning	245	56	1	192	89	21
Tanning	555	134	9	448	231	19

a.

	A/A	A/G	G/G	C/C	C/T	T/T	SUM	frequency
Burning	245	56	1	192	89	21	604	0.302
Tanning	555	134	9	448	231	19	1396	0.698

b.

	OCA2	MC1R		
total	2000	2000		
	A	G	C	T
numbers	1790	210	1600	400
allelic frequencies	0.895	0.105	0.8	0.2

c. d.

各基因型频率：

	A/A	A/G	G/G	C/C	C/T	T/T	SUM	frequency
Burning	245	56	1	192	89	21	604	0.302
Tanning	555	134	9	448	231	19	1396	0.698
SUM	800	190	10	640	320	40		
frequency	0.8	0.19	0.01	0.64	0.32	0.04		

总数（1000）乘两个频率得到 expected（E）值：

	A/A	A/G	G/G	C/C	C/T	T/T
Burning	241.6	57.4	3.0	193.3	96.6	12.1
Tanning	558.4	132.6	7.0	446.7	223.4	27.9

X^2=(O-E) ^2/E:

	A/A	A/G	G/G	C/C	C/T	T/T
Burning	0.05	0.03	1.35	0.01	0.60	6.59
Tanning	0.02	0.01	0.58	0.00	0.26	2.85

Total chi square:

	OCA2			MC1R		
	A/A	A/G	G/G	C/C	C/T	T/T
Burning	0.05	0.03	1.35	0.01	0.60	6.59
Tanning	0.02	0.01	0.58	0.00	0.26	2.85
total chi square	2.05			10.31		
df=2						
critical value=5.99	2.05<5.99			10.31>5.99		
	null hypothesis cannot be rejected			null hypothesis can be rejected		

No evidence for nonrandom association between burning and OCA2, while burning and MC1R are associated. Individuals having burning tendency have higher T/T frequency than expected, and because C/T meets expectation, T/T is an recessive trait.

e.

No. 可能是随机事件导致的结果，但不能说完全没有关系。